ModelIt!

Student Activity Pack

The Roller Coaster Equation

How Physics Turns Height Into Speed and Screams Into Science

Grade 8 | NGSS MS-PS3-1, MS-PS3-2 | Energy and Matter

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved.

Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

ModelIt!

What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What makes a roller coaster go fast without an engine pushing it?

2 Where on a roller coaster do you think you have the MOST energy?

3 What's the difference between potential energy and kinetic energy?

4 Draw an energy diagram for a ball rolling down a hill.

2 Scoring Rubric

ModelIt!

Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

ModelIt!

What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Coaster Height | Coaster Mass | Potential Energy | Kinetic Energy (Speed)

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

ModelIt!

Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

ModelIt!

Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: First Drop

Maximum height, normal mass

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Heavy Coaster

Maximum height, double mass

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Low Hill

Set height to 50% of maximum

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Friction — Energy lost to heat as the coaster moves along the track

Air Resistance — Energy lost to pushing through the air

G-Forces — The feeling of being pushed or pulled on curves and drops

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

ModelIt!

The Big Question

How does a roller coaster turn height into speed without an engine? Use evidence from your simulation to support your answers.

1 Why does the first hill of a roller coaster have to be the tallest?

2 If energy is conserved, why does a real roller coaster eventually stop?

3 How would a coaster on the Moon be different from one on Earth?

4 Why do you feel 'weightless' at the top of a hill but heavy at the bottom of a dip?

5 How is a roller coaster similar to a pendulum or a bouncing ball?

8 STEM Challenge: Design the Ultimate Roller Coaster

ModelIt!

YOUR ENGINEERING MISSION

Design a roller coaster that achieves the maximum thrill while obeying the laws of energy conservation.

THE SCENARIO

A theme park has hired your engineering team to design their new signature roller coaster. It must be thrilling but physically possible — every hill, loop, and turn must obey energy conservation.

Engineering Design Thinking

1. HOW TALL WILL YOUR FIRST HILL BE?

2. HOW MANY LOOPS AND HILLS WILL YOUR COASTER HAVE?

3. WHY MUST EACH SUBSEQUENT HILL BE SHORTER THAN THE PREVIOUS ONE?

4. HOW WILL YOU ACCOUNT FOR ENERGY LOST TO FRICTION?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Roller Coaster Engineers and Mechanical Engineers design rides using physics principles. They use computer models to test designs before building. They earn $75,000-$130,000/year.